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Multimodal-guided prototype calibration and temporal coherence-aware hybrid matching for few-shot action recognition

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ABSTRACT

Few-shot action recognition methods achieve impressive performance by learning discriminative features and designing temporal alignment strategies. However, these methods suffer from three significant challenges in distinguishing similar classes: (a) spatiotemporal and motion features are complementary and essential, yet the latter is frequently overlooked; (b) excessive reliance on unimodal video data leaves multimodal information underexplored, and (c) the temporal distribution of actions is prone to intra-class temporal offsets and inter-class local similarity. To overcome these challenges, we propose a multimodal-guided prototype calibration and temporal coherence-aware hybrid matching (MGTH), which integrates four innovative components: a motion-enhanced temporal aggregation module (MTAM), a text-guided prototype calibration module (TPCM), a video-text adapter objective (VTA), and a temporal coherence-aware hybrid matching (TCH). The MTAM encodes complementary spatiotemporal and motion features without any 3D convolution or optical flow calculations. The TPCM fully utilizes video-text information to optimize video prototypes. Meanwhile, the VTA maximizes the similarity between video features and corresponding textual representations. The TCH strategy alleviates metric bias caused by intra-class temporal offsets and inter-class local similarity through a hybrid matching mechanism, and strengthens this mechanism's ability to distinguish similar actions by applying temporal coherence regularization to the input video. Additionally, we extend the proposed MGTH to more challenging tasks, including cross-domain few-shot action recognition and zero-shot action recognition. Experimental results on multiple benchmark datasets demonstrate that MGTH achieves state-of-the-art performance, confirming the superiority of our approach.

Introduction

Over the past years, action recognition has made promising progress with the rapid development of large-scale datasets, deep neural networks (Qin and Liu, 2024, Yin et al., 2024), and Transformer models (Liu et al., 2023, Ahn et al., 2023, Wen et al., 2023). However, existing methods still face dual challenges regarding efficiency and generalization. On one hand, action recognition tasks typically rely on large-scale labeled data to ensure model performance. However, acquiring such data requires significant human resources and, in some cases, may be impossible, which limits the effectiveness of supervised learning methods. On the other hand, most research primarily focuses on “closed-set” classification problems, which assume that all action categories are predefined and visible to the model. Nevertheless, these methods encounter obstacles when handling unknown categories, especially in practical applications such as sports analytics (Bright et al.,

2024) and autonomous driving (Wang et al., 2023). In contrast, humans are capable of rapidly recognizing and understanding unseen actions with very few annotated examples. Importantly, few-shot learning (Wang et al., 2023, Lim et al., 2024, Li et al., 2023, Cao et al., 2020), as a transfer learning approach that closely resembles human learning processes, has emerged as a research hotspot across multiple fields.

Most few-shot learning methods primarily concentrate on image recognition; however, applying these techniques to action recognition tasks presents remarkable constraints. This is attributed to the fact that video possesses rich temporal information and contains complex scene changes and various action attributes. In this context, existing few-shot action recognition approaches can be broadly categorized into two types: data augmentation-based methods (Dwivedi et al., 2019, Zhang et al., 2020, Wu et al., 2022, Fu et al., 2020) and alignment-based methods (Cao et al., 2020, Bishay et al., 2019, Thatipelli et al., 2022, Wang et al., 2022). The former attempts to generate additional training

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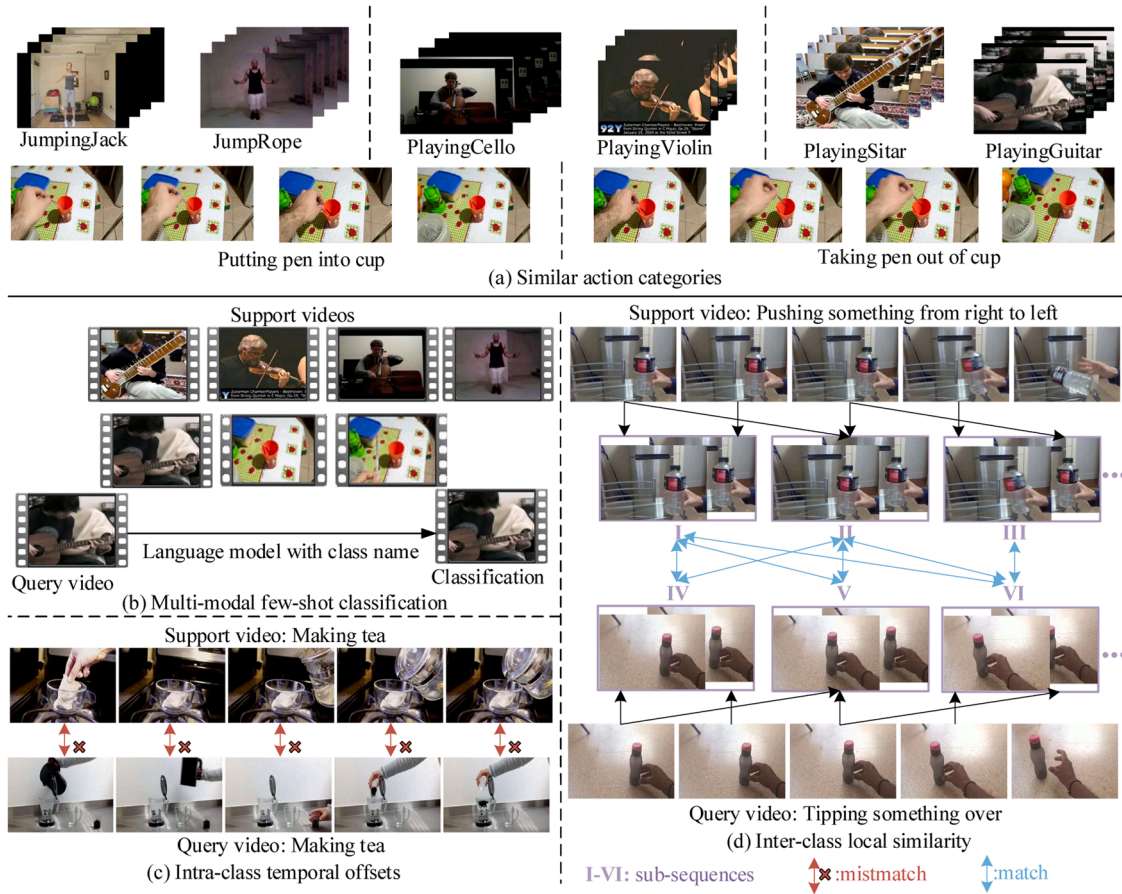


Fig. 1. Illustrates of our motivation. (a) When faced with a restricted number of samples, unimodal methods are susceptible to misclassifying categories that are similar in terms of scenes and sub-actions. However, in (b), the multimodal approach enhances the discrimination of feature representations by integrating visual and textual semantic information, resulting in more accurately distinguishing similar categories. Furthermore, many action instances exhibit intra-class temporal offsets and inter-class local similarity, which seriously impair the effectiveness of temporal alignment, as demonstrated in (b) and (c).

data (Dwivedi et al., 2019), self-supervised signals (Zhang et al., 2020), or auxiliary information (Wu et al., 2022, Fu et al., 2020) for robust representation learning. The latter measures the similarity between query and support samples using frame-level or segment-level alignment techniques. Despite the strides made in these studies, they mainly focus on limited unimodal video data, which are inadequate to comprehensively reflect the complex features of human actions, thereby impeding the accurate differentiation of similar categories. Taking the four sets of action categories in Fig. 1(a) (i.e., “JumpingJack” and “JumpRope”, “PlayingCello” and “PlayingViolin”, “PlayingSitar” and “PlayingGuitar”, “Putting pen into cup” and “Taking pen out of cup”) as instances, some existing methods (Cao et al., 2020) are prone to confusing these categories by assigning close prediction scores due to their similarity in scenes and sub-actions. Based on these observations, we analyze the underlying reasons from three perspectives.

The first challenge stems from underestimating the importance of motion features in action recognition (Lu et al., 2024, Wang et al., 2021, Li et al., 2020). As depicted in Fig. 1(a), distinguishing between “Putting pen into cup” and “Taking pen out of cup” is extremely difficult without motion information. Although spatiotemporal features can track the positional changes of objects, they often struggle to differentiate action types with similar spatial trajectories but in opposite directions. In contrast, motion information exhibits greater sensitivity to action direction by emphasizing the motion edges of adjacent frame features and suppressing their identical static appearance features. As demonstrated in many studies (Li et al., 2024, Wang et al., 2022), spatiotemporal and motion features are complementary in action recognition. However, existing few-shot methods mainly focus on spatiotemporal features and

put relatively little effort into acquiring motion information, resulting in poor performance.

The second challenge arises from the inability of most methods to effectively mitigate data scarcity, which restricts the accuracy of class prototypes and consequently leads to bias against unseen action categories. This is because these methods utilize only limited unimodal visual features for classification, ignoring information from additional modalities. In traditional action recognition, state-of-the-art methods employ multiple modalities (Yang et al., 2024, Yang et al., 2024, Wang et al., 2023) (e.g., visual, audio, and text) to provide complementary information for precise recognition. However, the potential of multi-modal information remains unexploited in few-shot action recognition tasks, while limited data further amplifies the deficiencies of unimodal approaches.

The third challenge involves possible intra-class temporal offsets and inter-class local similarity within the temporal distribution of actions, which constrain the performance of existing matching methods in handling misaligned videos. Specifically, literature (Cao et al., 2020) efficiently estimates ordered temporal alignment paths to match two video sequences. Nevertheless, certain categories manifest intra-class temporal offsets, characterized by frame offset phenomena between frame sequences sampled from the same category. As illustrated in Fig. 1(c), the water may be poured either before or after the tea leaves during the tea-making process. This discrepancy in order renders the aforementioned temporal alignment method ineffective. Accordingly, the study (Wang et al., 2022) proposes a bidirectional Mean Hausdorff Metric (Bi-MHM) to reduce the limitations of strictly ordered alignment and thereby improve the correspondence between the query and

support. However, it fails to capture temporal sequence information, leading to confusion among similar actions. To this end, Temporal-Relational Cross-Transformers (TRX) (Perrett et al., 2021) align each query subsequence with all subsequences in the support set, facilitating the matching of actions under varying speeds and temporal offsets to distinguish fine-grained categories. Due to the lack of long-range temporal relationships, the unordered matching of subsequences results in inter-class local similarity, where frames sampled from videos of different categories may exhibit similar motion patterns. As outlined in Fig. 1(d), in the context of subsequence matching of similar actions (V probably matches I and II), “Pushing something from right to left” and “Tipping something over” may be incorrectly recognized as the same category. This can be attributed to TRX’s excessive pursuit of aligning action subsequences, which neglects the inherent long-term temporal order cues of actions, resulting in suboptimal matching between query and support class prototypes. Therefore, another goal of few-shot action recognition is to design an effective matching strategy that better tackles various types of video tasks.

Based on the aforementioned considerations, we focus on exploiting multimodal knowledge for few-shot action recognition and thus propose a multimodal-guided prototype calibration and temporal coherence-aware hybrid matching (MGTH) framework. More specifically, to tackle the first challenge, inspired by (Wang et al., 2022, Wang et al., 2023), we design a motion-enhanced temporal aggregation module (MTAM) to encode complementary spatiotemporal and motion features. Presently, existing methods typically utilize optical flow (Wang et al., 2022, Piergiovanni and Ryoo, 2019) to integrate motion information, which inevitably incurs additional computational and storage costs. In contrast, the MTAM module adaptively extracts task-relevant motion features by calculating the difference between features obtained using learnable convolutions and those of adjacent video frames acquired without learnable convolutions. It is noteworthy that our objective is to identify a better and more lightweight method for obtaining motion representations that effectively recognize actions rather than relying on optical flow to capture precise motion information between two frames. Furthermore, the MTAM employs a Temporal Transformer to model global temporal relationships of spatial appearance features, resulting in spatiotemporal features. These spatiotemporal features are then combined with motion encodings in a straightforward manner.

To address the second challenge, we propose a text-guided prototype calibration module (TPCM) aimed at fully utilizing video-text multimodal information to explicitly modulate video prototypes. Due to the limited availability of annotated visual data and the absence of corresponding multimodal information, most existing methods (Cao et al., 2020, Bishay et al., 2019, Thatipelli et al., 2022, Perrett et al., 2021) fail to construct satisfactory prototypes. In response, we adopt a powerful emerging paradigm, contrastive language-image pre-training (CLIP) (Radford et al., 2021), as our baseline. This is because CLIP simultaneously encodes the input image and text and maps them into the same feature space, which enables cross-modal inference between image and text. Based on this property, the proposed TPCM module adaptively integrates bi-modal information from video (spatiotemporal information) and text (semantic information of class labels).

This process enriches the semantic representation of video prototypes and optimizes intra- and inter-class relevance. Additionally, to adapt the image-text matching model CLIP for video-text matching tasks, we design a video-text adapter objective (VTA) to minimize the distance between video features and their corresponding text features.

Considering the third challenge, we design a temporal coherence-aware hybrid matching (TCH) to alleviate intra-class temporal offsets and inter-class local similarity for efficiently tackling various few-shot action recognition tasks. Specifically, during frame-level matching, inspired by (Wang et al., 2022), we utilize the Bi-MHM metric to relax the strict ordering constraints between query and support samples. However, for actions strongly reliant on temporal information, such as “Putting something in the box” and “Taking something out of it,”

Bi-MHM may be limited due to its inability to capture temporal order. In contrast, drawing inspiration from TRX (Perrett et al., 2021), we observed that subsequences are constructed in strict temporal order, effectively compensating for the shortcomings of frame-level Bi-MHM. Accordingly, we propose a hybrid matching strategy that integrates frame-level and tuple-level Bi-MHM to mitigate intra-class temporal offsets and inter-class local similarity. However, this strategy focuses on short-term temporal patterns and is less effective in distinguishing complex actions that rely on long-range temporal relationships (e.g., “Pretending to pour something out of something but something is empty” and “pouring something into something”). To address this limitation, we introduce temporal coherence regularization into the input features, enhancing the model’s sensitivity to long-term temporal cues without adding extra parameters. By pulling adjacent frame features closer and pushing non-adjacent frame features farther apart, the method further strengthens the effectiveness of the hybrid matching strategy in complex scenarios.

In conclusion, we present the following four contributions:

- We propose a motion-enhanced temporal aggregation module to model spatiotemporal and motion features without relying on 3D convolution or optical flow computation, where the two types of features are treated as independent and complementary representations.
- We design a text-guided prototype calibration module that fully leverages the textual semantic priors from CLIP to compensate for the scarcity of video features and further refine video prototypes. Furthermore, to enhance the adaptability of CLIP in few-shot action recognition tasks, we construct a video-text adapter objective to maximize the similarity between video features and the corresponding textual representations.
- We propose a temporal coherence-aware hybrid matching approach that integrates frame-level and tuple-level matching strategies to mitigate metric bias caused by intra-class temporal offsets and inter-class local similarity. Meanwhile, the method applies temporal coherence regularization to the input video to explicitly exploit long-range temporal information, thereby further enhancing the discriminative ability of the matching strategy when distinguishing highly similar actions.
- Extensive experiments on four benchmarks, HMDB-51, UCF-101, Kinetics, and SSv2, demonstrate the effectiveness and rationale of the proposed MGTH model compared to the state-of-the-art methods.

Related work

Action recognition

Video action recognition emerges as a significant research topic, driven by the increasing demand for video analysis in real-world applications. Early studies primarily rely on handcrafted feature extraction methods. For example, Literature (Vishwakarma and Dhiman, 2019) extracts human silhouettes through entropy-based texture segmentation and constructs average energy silhouette images (AEIs) to reduce the time complexity of feature generation. Similarly, this work (Dhiman and Vishwakarma, 2019) constructs robust feature vectors by computing R-transform features and Zernike moments on AESIs. With the rapid progress of deep learning, action recognition performance improves significantly. Some works employ 2D convolutional neural networks (2D CNNs) for video action recognition. For instance, ACTION-Net (Wang et al., 2021) utilizes multipath excitation mechanisms to capture spatiotemporal, channel-wise, and motion features. However, 2D CNNs inherently lack the capacity to model temporal sequences, whereas temporal information plays a crucial role in action recognition. To address this limitation, some approaches (Dhiman and Vishwakarma, 2020, Dhiman et al., 2021) utilize Bi-LSTM and LSTM to learn temporal dynamic characteristics. In addition, many methods employ 3D

convolutional neural networks (3D CNNs) to directly learn spatiotemporal representations. For example, X3D (Feichtenhofer, 2020) achieves an effective balance between efficiency and accuracy, while SlowFast (Qi et al., 2020) captures motion information through the coordinated modeling of slow and fast pathways. However, 3D CNN-based methods typically require substantially higher computational costs. Consequently, several studies such as TDN (Wang et al., 2021) combine 2D spatial convolutions with 1D temporal convolutions as a lightweight alternative to full 3D convolution. In recent years, Transformer-based approaches for video action recognition (Liu, 2022, Yang et al., 2022) attract increasing attention and achieve performance that surpasses convolutional networks on multiple benchmark datasets. However, these video action recognition methods rely heavily on large amounts of annotated data, while collecting and labeling high-quality video samples is often costly and impractical. Motivated by these challenges, this work focuses on the problem of few-shot action recognition, which aims to accurately recognize novel action categories using only a few labeled samples.

Vision-language contrastive learning

Fundamentally, the same event or object can be represented using data from various modalities. For instance, an image commonly characterized by visual features can also be described with semantic features extracted from corresponding textual information. Recently, vision-language contrastive pre-trained models (Jia et al., 2021, Zhong et al., 2022) have received increasing attention by exploiting associations among multimodal data. Among them, the most representative work, CLIP (Radford et al., 2021), utilizes two encoders to project images and text into a shared feature space for contrastive learning and is pre-trained on large-scale image-text pairs to achieve superior zero-shot capability. Therefore, a large number of CLIP-based studies leverage pre-trained models to improve performance in downstream tasks, such as image classification (Zhou et al., 2022) and object detection (Song and Bang, 2024). However, directly applying pre-trained CLIP models to videos is extremely difficult due to the absence of video-specific temporal cues in image-level pre-training. To address this issue, several works (Wang et al., 2023, Chen et al., 2022, Ni et al., 2022) adapt CLIP for video by integrating additional learnable components such as self-attention layers, textual or visual prompts, or specialized visual decoders. Ju et al. (Chen et al., 2022) perform temporal modeling of CLIP using text prompts and transformer layers. Nevertheless, this approach limits the generalization of CLIP and leads to suboptimal performance in the zero-shot setting. ActionCLIP (Wang et al., 2023) provides a new perspective on action recognition by treating action recognition as a video-text multimodal matching problem. X-CLIP (Ni et al., 2022) utilizes a cross-frame attention mechanism to capture long-range inter-frame dependencies in video and generates discriminative textual prompts using video information. Additionally, Text4Vis (Wu et al., 2023) and ViFi-CLIP (Rasheed et al., 2023) significantly enhance model performance through fine-tuning CLIP. However, the aforementioned method overlooks motion features that complement the visual information missed by normal video frames. To this end, we propose a CLIP-based motion-enhanced temporal aggregation module that models the global temporal relationships of spatial features and encodes motion features between adjacent frame representations.

Motion dynamics

Motion dynamics have been proven to provide crucial insights for action recognition. Recent few-shot action recognition studies (Dwivedi et al., 2019, Zhang et al., 2020, Bishay et al., 2019) leverage 3D CNNs to model both appearance and motion features simultaneously. However, 3D CNN-based methods require extensive parameter optimization, making it difficult to achieve an effective balance between accuracy and speed. Additionally, some works based on two-stream networks

(Simonyan and Zisserman, 2014, Wang et al., 2016), primarily designed for traditional action recognition, represent motion information in videos using optical flow. These approaches inevitably increase computational complexity and storage requirements. To solve this problem, several approaches (Li et al., 2020, Wang et al., 2022, Jiang et al., 2019) develop temporal difference modules that can be integrated into 2D CNNs to learn feature-level motion representations. Inspired by (Jiang et al., 2019), we interpret temporal differences as motion features and incorporate them into the field of few-shot action recognition.

Few-shot action recognition

The core concept of few-shot action recognition is similar to that of few-shot image recognition, but it is more complex due to the additional temporal dimension. Currently, most existing methods (Cao et al., 2020, Dwivedi et al., 2019, Zhang et al., 2020, Wu et al., 2022, Fu et al., 2020, Bishay et al., 2019, Thatipelli et al., 2022, Wang et al., 2022, Perrett et al., 2021, Dhiman and Vishwakarma, 2019, Xing et al., 2023, Zhang et al., 2021, Huang et al., 2022, Nguyen et al., 2022, Wang et al., 2024, Yang et al., 2023) utilize a metric-based learning strategy to measure the similarity between videos in a unified metric or feature space, facilitating few-shot inference. For instance, Cao et al. (Cao et al., 2020) design a temporal alignment method to match frames between query and support videos. Perrett et al. (Perrett et al., 2021) utilize a Cross-Transformer attention mechanism to match each query subsequence with all support subsequences, achieving optimal 5-way 5-shot action recognition performance. Literature (Xing et al., 2023) employs long- and short-term temporal modeling modules to extract enriched spatiotemporal features, which are then fed into a matching method similar to TRX (Dhiman and Vishwakarma, 2019) for few-shot action recognition. Another work (Thatipelli et al., 2022) applies local patch-level enrichment and global frame-level enrichment sub-modules to enhance spatiotemporal feature representations and utilizes these representations to learn the relational matching between query and support action subsequences. Zhang et al. (Zhang et al., 2021) propose a frame-wise implicit temporal alignment method to achieve robust video matching. The studies (Wu et al., 2022, Wang et al., 2022) indicate the importance of learning task-specific features. Huang et al. (Huang et al., 2022) and Nguyen et al. (Nguyen et al., 2022) utilize multiple similarity functions for accurate video frame matching. Inspired by data augmentation, some methods attempt to introduce additional self-supervised signals to learn robust video representations (Zhang et al., 2020) or employ generative adversarial networks (Dwivedi et al., 2019, Wang et al., 2024) to synthesize more examples of new classes. Additionally, Fu et al. (Fu et al., 2020) leverage depth information as a carrier of the scene to train action models alongside RGB data. Meanwhile, AMFAR (Yang et al., 2023) actively selects reliable modalities for each sample based on task-dependent contextual information to enhance few-shot inference. Unlike the above methods, we propose an MGTH model to effectively utilize the strong multimodal knowledge of CLIP and further investigate the relationship between textual and visual modalities.

Method

Preliminaries of CLIP

Unlike the traditional supervised training paradigm that relies on labeled samples, CLIP leverages text-driven semantic features to guide representation learning, thereby enhancing the model's transferability and scalability. Specifically, CLIP employs a text encoder (Transformer) and a visual encoder (ResNet or ViT) to learn robust and generalizable features from both text and images. The Information Noise-Contrastive Estimation (InfoNCE) technique then optimizes the parameters of these two encoders through contrastive learning on the extracted image and text features. The objective is to pull them closer if these two types of features match; otherwise, push them apart. To ensure consistency

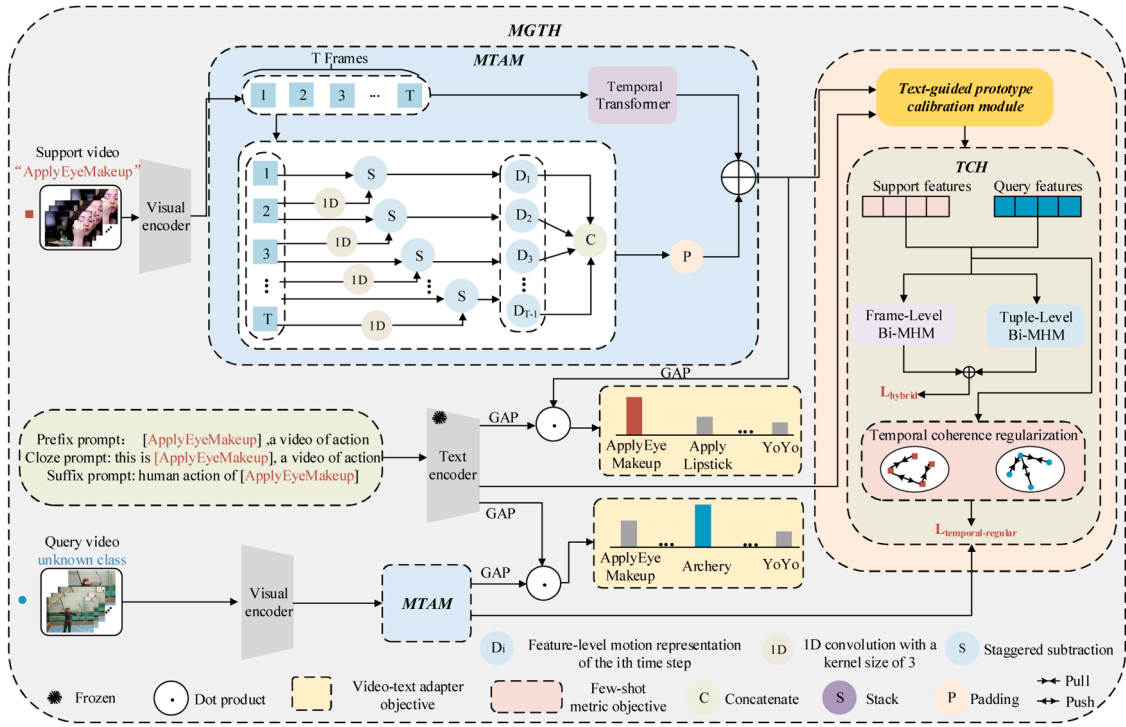


Fig. 2. Overview of MGTH. Given the support and query videos, visual and text features are extracted using CLIP’s visual and text encoders. Next, a motion-enhanced temporal aggregation module (MTAM) encodes complementary spatiotemporal and motion features. Subsequently, a video-text adapter objective (VTA) migrates the image-text matching model (i.e., CLIP) to the video-text matching task. Afterward, a text-guided prototype calibration module (TPCM) integrates visual and textual features to refine the visual prototypes. Finally, a temporal coherence-aware hybrid matching (TCH) evaluates the similarity between the calibrated prototypes and query features for the few-shot metric. For clarity, the figure does not show other support and query videos in the few-shot task.

with the textual descriptions during pre-training, the input text conventionally utilizes the prompt template “a photo of [CLS]”, where [CLS] stands for the actual category name.

Problem formulation of few-shot action recognition

In few-shot classification, the dataset is typically divided into three subsets: a training set D_{train} , a validation set D_{val} , and a test set D_{test} . Importantly, the label spaces of these datasets are distinct and non-overlapping. Following established methodologies (Cao et al., 2020, Perrett et al., 2021), we employ episode-based meta-learning strategies to optimize network models, where each episode comprises a support set S and a query set Q . The support set contains M categories, each with K -labeled samples. Furthermore, the query set includes unlabeled samples from the same M categories. Few-shot models aim to predict the labels of the query set under the supervision of the support set. During inference, many episodic tasks are randomly selected from the D_{test} and the average accuracy is computed to assess the effectiveness of the proposed model.

Architecture overview

In Fig. 2, we present the framework of the proposed MGTH using an M -way 1-shot task as an illustrative example. Specifically, each input video with variable length is divided into T equal segments following the method in previous works (Cao et al., 2020, Wang et al., 2022), and one frame is randomly sampled from each segment. Accordingly, in the episodic task, the support set $S = \{s_1, s_2, s_3, \dots, s_M\}$ consists of M video samples, where $s_i \in \mathbb{R}^{T \times 3 \times H \times W}$, with T denoting the number of frames per video, and H and W representing the spatial dimensions. Importantly, the query video Q comprises similar elements to the support set S . Next, the visual encoder V of CLIP is used to extract feature representations of the input video frames, while the text encoder W generates

textual features corresponding to class language descriptions. Notably, compared to “a photo of [CLS]”, the adopted class text description contains three types of prompts: Prefix prompt, Cloze prompt, and Suffix prompt. These diverse semantic cues enable the model to more accurately recognize unseen action categories. Then, we propose a motion-enhanced temporal aggregation module (MTAM), which aims to extract motion information from adjacent frame differences and derive spatiotemporal features by employing a Temporal Transformer for global temporal modeling of spatial features. To adapt CLIP for few-shot action recognition tasks, we construct a video-text adapter objective (VTA) to maximize the similarity between the video features and their corresponding textual class representations. Additionally, we design a text-guided prototype calibration module (TPCM) that fully leverages video-text multimodal information to enhance the representativeness of video prototypes. Finally, the calibrated prototypes are measured with query features using the proposed temporal coherence-aware hybrid matching (TCH) to achieve robust few-shot recognition. Further details are discussed in Sections 3.4 to 3.8.

Motion-enhanced temporal aggregation module

In action recognition tasks, spatial features extracted from video frames fail to perceive motion and temporal information. However, spatiotemporal and motion features are complementary and crucial for this task. Consequently, we propose a motion-enhanced temporal aggregation module (MTAM) to integrate these two features and thus generate a more comprehensive video-level representation.

Given a support set $S = \{s_1, s_2, s_3, \dots, s_M\}$ and a query set $Q = \{q_1, q_2, q_3, \dots, q_M\}$, where $\{s_i, q_i\} \in \mathbb{R}^{T \times 3 \times H \times W}$. The visual encoder V is used to extract spatial features for each video frame:

$$H_{s_i} = V(s_i), H_{q_i} = V(q_i) \quad (1)$$

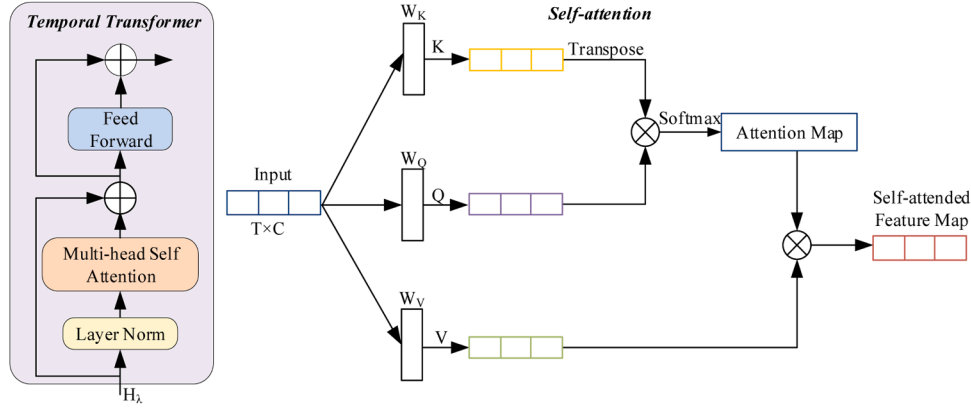


Fig. 3. Architecture of the Temporal Transformer.

where $\{H_{s_i}, H_{q_i}\} \in \mathbb{R}^{T \times C}$, and C represents the dimension of the frame feature. We then employ a Temporal Transformer module to model long-term temporal relations on the extracted spatial appearance, as shown in Fig. 3. Specifically, H_{s_i} and H_{q_i} perform multi-head self-attention along the temporal dimension to aggregate global temporal information. Afterward, residual learning is employed to ensure more stable model convergence.

$$\widetilde{H}_\lambda = (\text{Module}_{att}(\text{LN}(H_\lambda))) + H_\lambda \quad (2)$$

where $\lambda \in \{s_1, s_2, \dots, s_M, q_1, q_2, \dots, q_M\}$, $\widetilde{H}_\lambda \in \mathbb{R}^{T \times C}$, LN represents layer normalization, and Module_{att} denotes a multi-head self-attention mechanism. For simplicity, we set the number of attention heads in this mechanism to one, denoted as $\text{Module}_{att}^{(1)}$. The self-attention module first applies three learnable linear projection matrices, W_K , W_Q , and W_V , to the input feature H_λ to obtain the Query (Q), Key (K), and Value (V) representations. The matrix K is then transposed and multiplied with Q to compute the attention scores. After normalization with the Softmax function to produce the attention weight map, these weights are finally multiplied with V to generate the output features.

$$Q = W_Q H_\lambda, K = W_K H_\lambda, V = W_V H_\lambda \quad (3)$$

$$\text{Module}_{att}^{(1)}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{C_d}}\right) V \quad (4)$$

where $\frac{1}{\sqrt{C_d}}$ is the scaling factor. Subsequently, a residual feed-forward network (FFN) is employed to refine feature pointwise, as given by:

$$\widetilde{H}_\lambda' = \text{FFN}(\widetilde{H}_\lambda) + \widetilde{H}_\lambda \quad (5)$$

where the FFN comprises a series of fully connected layers, GELU, and Dropout layers.

Additionally, the motion representation at time step t is

approximately defined as the difference between adjacent frames $H_\lambda(t)$ and $H_\lambda(t+1)$. It is noteworthy that we apply a 1D convolution with a kernel size of 3, stride of 1, and padding of 1 along the channel dimension of $H_\lambda(t+1)$, which enables efficient local cross-channel interactions and facilitates the learning of task-relevant features. This process is formulated as follows:

$$D_\lambda(t) = \text{Conv}_{1D} * H_\lambda(t+1) - H_\lambda(t) \quad (6)$$

where $t \in \{1, 2, 3, \dots, T-1\}$, $D_\lambda(t) \in \mathbb{R}^C$ indicates the feature-level motion representation at time step t . Next, all motion features are concatenated along the temporal dimension to construct the final motion matrices, denoted as $D_\lambda \in \mathbb{R}^{(T-1) \times C}$.

To maintain the same temporal dimension as $\widetilde{H}_\lambda'$, we use zeros to represent the feature information of the final time step in the motion feature, i.e., $D_\lambda(T) = 0$, resulting in $D_\lambda' \in \mathbb{R}^{T \times C}$, as depicted in Algorithm 1. Finally, a weighted sum is applied to the spatiotemporal features $\widetilde{H}_\lambda'$ and motion features D_λ' . This process is encapsulated by the following equation:

$$F_\lambda = \widetilde{H}_\lambda' + D_\lambda' \quad (7)$$

Video-text adapter objective

Existing approaches employ a text encoder T to extract class text features with the prompt template “a photo of [CLS]”. However, the lack of diversity in a single textual prompt limits the adaptation of upstream pre-trained models to downstream tasks. Inspired by the ActionCLIP (Wang et al., 2023) model, we introduce three types of templates—Prefix prompt, Cloze prompt, and Suffix prompt—to enrich the contextual information of the text. This approach effectively enhances the model’s reliability in handling various video tasks, enabling it to more accurately understand and respond to complex contexts and dynamic changes within video content. After that, the obtained text features are denoted as $\{Z_i\}_{i=1}^K$, where $Z_i \in \mathbb{R}^{P \times C}$, K represents the number of classes in the training set D_{train} , and P stands for the number of text prompt templates. To retain CLIP’s rich textual semantic priors and alleviate the optimization workload, we freeze the weights of the encoder T during training. In contrast, the visual encoder V is fine-tuned to achieve better alignment between visual and textual features.

To migrate the image-text matching model CLIP to the video-text matching task, we optimize the similarity between video and text features by simulating CLIP’s original training objective. More precisely, the similarity is maximized if the video and text are matched and minimized otherwise. To reach this objective, we design a video-text adapter objective to more comprehensively compute the video-text matching probability, as illustrated in Equations (8) and (9).

Algorithm 1

Motion encoding in MTAM.

Input: Frame-level features $\{H_\lambda(1), H_\lambda(2), H_\lambda(3), \dots, H_\lambda(T)\}$ extracted by the visual encoder
Output: $D_\lambda \in \mathbb{R}^{T \times C}$
Motion feature: $D_\lambda = \text{List}()$
for $t = 1$ **to** $T - 1$ **do**
 $D_\lambda(t) = \text{Conv}_{1D} * H_\lambda(t+1) - H_\lambda(t)$
 D_λ . Append ($D_\lambda(t)$)
end
 D_λ . Append (0)
 $D_\lambda = \text{Stack}(D_\lambda)$

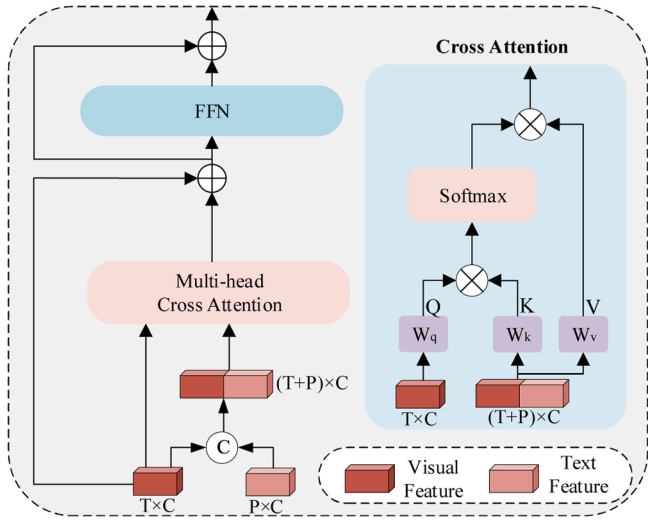


Fig. 4. The overall structure of the proposed TPCM.

$$P_{\text{video-text}} = \frac{\exp(\text{sim}(\text{GAP}(F_{\lambda}), \text{GAP}(Z_i))/\tau)}{\sum_{i=1}^K \exp(\text{sim}(\text{GAP}(F_{\lambda}), \text{GAP}(Z_i))/\tau)} \quad (8)$$

$$P_{\text{text-video}} = \frac{\exp(\text{sim}(\text{GAP}(Z_i), \text{GAP}(F_{\lambda}))/\tau)}{\sum_{\lambda=1}^{2M} \exp(\text{sim}(\text{GAP}(Z_i), \text{GAP}(F_{\lambda}))/\tau)} \quad (9)$$

Here, sim denotes the cosine similarity function, and GAP serves as the global average pooling layer used to obtain video and text prompt representations. τ represents a learnable temperature parameter. Cross-entropy loss functions are then applied separately to $P_{\text{video-text}}$ and $P_{\text{text-video}}$, and these two losses are combined to achieve cross-modal feature alignment, thereby enhancing the model's generalization and robustness.

Text-guided prototype calibration module

In few-shot tasks, visual information is often insufficient due to data scarcity, resulting in less accurate class prototypes. Fortunately, the visual and textual modalities are complementary, and CLIP contains strong textual semantic priors. Accordingly, inspired by (Han et al., 2024), we design a text-guided prototype calibration module (TPCM) that utilizes the transferable textual knowledge from CLIP to improve the representativeness of video prototypes, as illustrated in Fig. 4. For simplicity, the number of attention heads is set to 1 to demonstrate the cross-attention mechanism.

Specifically, within the support set, the visual feature F_{s_i} is used to compute the visual prototype F_{s_i}' through the mean-of-class paradigm. After that, F_{s_i}' and the corresponding class text features Z_i are concatenated, and the resulting features are linearly transformed into K and V . Additionally, the visual prototype F_{s_i}' is transformed to form Q , which is then employed in attention learning with the K and V . This process is formulated as follows:

$$Q = W_q F_{s_i}', K = W_k [F_{s_i}', Z_i], V = W_v [F_{s_i}', Z_i] \quad (10)$$

where $[]$ denotes the concatenation operation. Subsequently, the scaled dot-product attention is performed as defined:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{C_d}}\right)V \quad (11)$$

Next, a residual connection and an FFN module are applied:

$$V_{s_i} = \text{FFN}(\text{Attention}(Q, K, V) + F_{s_i}') \quad (12)$$

In summary, TPCM employs a multi-head cross-attention mechanism

to adaptively associate the semantic information of labeled text with the corresponding video features, which significantly enhances the reliability of the video prototype.

Temporal coherence-aware hybrid matching

Human actions are typically complex, comprising numerous sub-actions with varying orders and offsets, which makes it challenging for existing matching methods (Cao et al., 2020) to effectively address intra-class temporal offsets. To this end, HyRSM (Wang et al., 2022) employs a Bi-MHM metric to mitigate strict ordering constraints, resulting in better correspondence between query and support samples. However, Bi-MHM fails to capture temporal order information, which may cause confusion when distinguishing actions that are highly sensitive to temporal order, such as “pick up a glass of water” and “put down a glass of water.” To overcome this limitation, subsequent work (Perrett et al., 2021) groups multiple sequential frames into tuples to preserve the temporal structure of actions, thereby compensating for the shortcomings of frame-level Bi-MHM matching. However, due to the fixed-length frame combination mechanism, tuples struggle to capture long-term temporal information within action sequences, ultimately resulting in suboptimal estimation of action distances. To alleviate intra-class temporal offsets and inter-class local similarity, we propose a temporal coherence-aware hybrid matching (TCH) method. This approach is composed of a hybrid attention mechanism that integrates frame-level and tuple-level Bi-MHM, together with temporal coherence regularization, aiming to enhance the model's discriminative capability across various few-shot action recognition tasks. The details are as follows:

Frame-level Bi-MHM Given the task-oriented features F_{s_i} and F_{q_i} , the k -th prototype video feature in the m class and the p -th query video feature are denoted as $p_m^k \in \mathbb{R}^{T \times C}$ and $q_p \in \mathbb{R}^{T \times C}$, respectively. For single-frame matching, we utilize the Bi-MHM metric, defined as follows:

$$D_{\text{frame-level}} = \frac{1}{T} \left[\sum_{i=1}^T \left(\min_{q_{p_j} \in q_p} \left\| p_m^{k,i} - q_{p_j} \right\| \right) + \sum_{j=1}^T \left(\min_{p_m^{k,i} \in p_m^k} \left\| q_{p_j} - p_m^{k,i} \right\| \right) \right] \quad (13)$$

Here, $\|\cdot\|$ represents the cosine distance, $p_m^{k,i}$ denotes the i -th video frame feature of the prototype p_m^k , and q_{p_j} stands for the j -th video frame feature of the query q_p .

Tuple-level Bi-MHM Frame-level Bi-MHM assumes that different videos of the same action class exhibit a similar distribution structure in feature space, but do not explicitly leverage temporal order information. Inspired by TRX (Perrett et al., 2021), we incorporate temporal relationships among multiple frames into the matching process to enhance recognition performance. To this end, we define the cardinality set as Ω . When $\Omega = \{2\}$, the action is represented through tuples formed by ordered frame pairs, as detailed in Equations (14) and (15).

$$\Gamma p_m^{k,i} = [p_m^{k,i_1} + PE(i_1), p_m^{k,i_2} + PE(i_2)] \quad (14)$$

$$\Gamma q_{p,j} = [q_{p,j_1} + PE(j_1), q_{p,j_2} + PE(j_2)] \quad (15)$$

where $1 \leq i_1 < i_2 \leq T$, $1 \leq j_1 < j_2 \leq T$, $\{p_m^{k,i_1}, q_{p,j_1}\} \in \mathbb{R}^{2C}$, $\Gamma p_m^{k,i}$ is the i -th frame pair representation of the support prototype, $\Gamma q_{p,j}$ is the j -th frame pair representation of the query video, and PE represents the positional encoding of frame index. Next, $\Gamma q_{p,j}$ is compared with all frame pair representations $\{\Gamma p_m^{k,i}\}_{i=1}^L$ from the support prototype using cosine distance, where $L = \frac{1}{2}T(T-1)$ denotes all possible frame pair combinations. Similarly, the support prototype is compared with all frame pair information in the query video. Consequently, the tuple-based Bi-MHM

matching strategy is defined as follows:

$$D_{\text{tuple-level}} = \frac{1}{L} \left[\sum_{i=1, \Gamma p_m^{k,i} \in \Gamma q_p}^L \left(\min_{\Gamma q_{p,j} \in \Gamma q_p} \|\Gamma p_m^{k,i} - \Gamma q_{p,j}\| \right) + \sum_{j=1, \Gamma q_{p,j} \in \Gamma q_p}^L \left(\min_{\Gamma p_m^{k,i} \in \Gamma p_m^k} \|\Gamma q_{p,j} - \Gamma p_m^{k,i}\| \right) \right] \quad (16)$$

Finally, the hybrid matching consisting of tuple-level and frame-level Bi-MHM is described:

$$D_{\text{hybrid}} = D_{\text{frame-level}} + D_{\text{tuple-level}} \quad (17)$$

Temporal coherence regularization We introduce temporal coherence regularization on the input video to enhance the network's ability to capture long-term temporal order cues, rather than being limited to temporal relationships within fixed-length frame combinations. This regularization method pulls adjacent frames closer and pushes non-adjacent frames farther apart, effectively handling various types of video tasks without requiring additional network parameters, as defined in:

$$I(p_m^k; p_m^{k,a}, p_m^{k,b}) = \begin{cases} \frac{1}{(a-b)^2 + 1} \cdot \|p_m^{k,a} - p_m^{k,b}\|, & \text{if } |a-b| \leq \delta \\ \max(0, m_{ab} - \|p_m^{k,a} - p_m^{k,b}\|) & \text{if } |a-b| > \delta \end{cases} \quad (18)$$

where $p_m^{k,a}$ and $p_m^{k,b}$ are the a -th and b -th frame features of the prototype video p_m^k , respectively, and $m_{ab} = 1 - e^{-\frac{(|a-b|-\delta)^2}{2\sigma^2}}$ denotes the margin size to ensure smooth temporal coherence. Here, δ denotes the window size and is treated as a hyper-parameter, set to 1 for the HMDB-51 and SSv2 datasets and 3 for the Kinetics and UCF-101 datasets. Moreover, σ is a hyper-parameter controlling the influence of temporal distance in the margin, with values of 51 for HMDB-51 and SSv2 and 23 for Kinetics and UCF-101 in our experiments. Similar to $I(p_m^k; p_m^{k,a}, p_m^{k,b})$, we also apply temporal coherence regularization to the query video frames. Next, the temporal coherence regularization loss for both the prototype and query videos is defined as:

$$L_{\text{temporal-regular}} = \frac{1}{L} \left[\sum_{a=1}^T \sum_{b=1, a \neq b}^T \left(\left(\lambda \cdot \frac{1}{(a-b)^2 + 1} \cdot \|p_m^{k,a} - p_m^{k,b}\| + \alpha \cdot (1-\lambda) \cdot \max(0, m_{ab} - \|p_m^{k,a} - p_m^{k,b}\|) \right) + \left(\lambda \cdot \frac{1}{(a-b)^2 + 1} \cdot \|q_{p,a} - q_{p,b}\| + \alpha \cdot (1-\lambda) \cdot \max(0, m_{ab} - \|q_{p,a} - q_{p,b}\|) \right) \right) \right] \quad (19)$$

where $q_{p,a}$ and $q_{p,b}$ represent the a -th and b -th frame features of the query video q_p , respectively, and α is used to balance the loss between adjacent and non-adjacent frames. If $|a-b| \leq \delta$, then $\lambda = 1$; otherwise, $\lambda = 0$.

In a word, the proposed TCH strategy reduces inter-class local similarity and intra-class temporal offsets caused by similar video frames or varying frame orders, resulting in a more flexible matching metric.

Final loss

During model training, we employ cross-entropy losses based on D_{hybrid} , $P_{\text{video-text}}$, and $P_{\text{text-video}}$, which are denoted as L_{hybrid} , $L_{\text{video-text}}$, and $L_{\text{text-video}}$, respectively. In addition, a temporal coherence regularization loss $L_{\text{temporal-regular}}$ is introduced to jointly optimize the model parameters. Among these components, $L_{\text{video-text}}$ and $L_{\text{text-video}}$ maximize the similarity between video features and their corresponding textual features, thereby facilitating the effective transfer of CLIP from image-text

matching to video-text matching tasks. The L_{hybrid} is employed to improve query-support correspondence, mitigating intra-class temporal offsets and inter-class local similarity issues. However, L_{hybrid} primarily focuses on short-term temporal patterns, leading to reduced discriminative power for complex actions that rely on long-range temporal dependencies. To address this limitation, $L_{\text{temporal-regular}}$ enhances the model's sensitivity to long-term temporal structures by pulling together adjacent frame features while pushing apart non-adjacent ones, thereby improving the effectiveness of the hybrid matching strategy across various few-shot action recognition tasks. This process can be formulated as follows:

$$L = L_{\text{hybrid}} + \beta L_{\text{temporal-regular}} + \frac{(1-\beta)}{2} (L_{\text{video-text}} + L_{\text{text-video}}) \quad (20)$$

where β is a balance factor.

Experiments

Datasets

We conduct extensive experiments to evaluate the performance of the proposed MGTH on the following four challenging few-shot action recognition benchmarks.

- (1) **Kinetics**: The Kinetics dataset (Wang et al., 2022) consists of 400 action classes with 306,245 videos. Following the methodology of CMN (Huang et al., 2022), 100 action classes are sampled from this dataset to create Kinetics-100 for few-shot action recognition. The dataset includes 64 categories with 6,389 videos for training, 12 categories with 1,199 videos for validation, and 24 categories with 2,395 videos for testing.
- (2) **UCF-101**: This dataset (Soomro et al., 2012) comprises 13,320 videos spanning 101 action categories. Following the splitting strategy described in (Wang et al., 2024), we use 70 categories with 9,154 videos for training, 10 categories with 1,421 videos for validation, and 21 categories with 2,745 videos for testing.
- (3) **HMDB-51**: The HMDB-51 dataset (Kuehne et al., 2011) contains 6,849 videos from 51 action classes. We adopt the splitting

method proposed by TRX (Perrett et al., 2021), assigning 31 categories with 4,280 videos for training, 10 categories with 1,194 videos for validation, and 10 categories with 1,292 videos for testing.

- (4) **SSv2**: The SSv2 dataset (Goyal et al., 2017) contains 220,847 videos across 174 action classes. We utilize the splitting setting presented in (Cao et al., 2020), selecting 64 categories with 77,500 videos for training, 12 categories with 1,925 videos for validation, and 24 categories with 2,854 videos for testing.

Implementation details

For all datasets, following the previous studies (Wang et al., 2022, Perrett et al., 2021), we uniformly sparsely sample 8 frames from each video and resize the extracted frames to 256×256 . During training, data augmentation is performed using horizontal flipping and random cropping. However, many categories in the fine-grained SSv2 dataset are

Table 1

Comparison with state-of-the-art methods on the HMDB-51 and UCF-101 datasets. The classification accuracy results for the 5-way 1-shot and 5-way 5-shot settings are illustrated.

Modality	Model	Backbone	HMDB-51		UCF-101	
			1-shot	5-shot	1-shot	5-shot
Unimodal	OTAM (Cao et al., 2020)	ResNet-50	54.5%	68.0%	79.9%	88.9%
	TA2N (Li et al., 2021)	ResNet-50	57.1%	74.0%	80.9%	93.2%
	TRX (Perrett et al., 2021)	ResNet-50	53.1%	75.6%	78.2%	96.1%
	HyRSM (Wang et al., 2022)	ResNet-50	60.3%	76.0%	83.9%	94.7%
	HyRSM++ (Wang et al., 2022)	ResNet-50	61.5%	76.4%	85.8%	95.9%
	TTAN (Li et al., 2021)	ResNet-50	57.1%	74.0%	80.9%	93.2%
	STRM (Thatipelli et al., 2022)	ResNet-50	52.3%	77.3%	80.5%	96.9%
	HCL (Zheng et al., 2022)	ResNet-50	59.1%	76.3%	82.5%	93.9%
	VIM (Liu et al., 2023)	ResNet-50	61.1%	76.5%	86.2%	96.1%
	GgHM (Xing et al., 2023)	ResNet-50	61.2%	76.9%	85.2%	96.3%
	TADRNet (Wang et al., 2023)	ResNet-50	64.3%	78.2%	86.7%	96.4%
	Huang et al. (Huang et al., 2022)	ResNet-50	60.1%	77.0%	71.4%	91.0%
	Nguyen et al. (Nguyen et al., 2022)	ResNet-50	59.6%	76.9%	84.9%	95.9%
	MTFAN (Wu et al., 2022)	ResNet-50	59.0%	74.6%	84.8%	95.1%
	MTE (Jiang et al., 2026)	ResNet-50	61.8%	77.9%	86.6%	96.2%
	HM2 (Gao et al., 2025)	ResNet-50	60.3%	78.3%	86.3%	97.4%
	MSVT (Chen et al., 2025)	ResNet-50	62.1%	78.7%	90.4%	97.7%
Multimodal	AMeFu-Net (Fu et al., 2020)	ResNet-50	60.2%	75.5%	85.1%	95.5%
	SRPN (Wang et al., 2021)	ResNet-50	61.6%	76.2%	86.5%	95.8%
	CCLN (Wang et al., 2024)	ResNet-50	65.1%	78.8%	86.9%	96.1%
	CLIP-CPM ² C (Guo et al., 2025)	CLIP-ResNet50	66.3%	81.2%	91.1%	97.4%
	CLIP-MDMF (Guo et al., 2024)	CLIP-ResNet50	66.8%	83.0%	94.3%	98.8%
	MA-CLIP (Xing et al., 2025)	CLIP-ResNet50	73.3%	82.1%	91.8%	96.6%
	CLIP-FSAR (Wang et al., 2024)	CLIP-ResNet50	69.2%	80.3%	91.3%	97.0%
	MGTH	CLIP-ResNet50	73.9%	82.7%	91.8%	97.6%
	CLIP-CPM ² C (Guo et al., 2025)	CLIP-ViT-B	75.9%	88.0%	95.0%	98.6%
	CLIP-MDMF (Guo et al., 2024)	CLIP-ViT-B	77.1%	88.0%	97.0%	99.3%
	MA-CLIP (Xing et al., 2025)	CLIP-ViT-B	83.4%	87.9%	96.5%	98.6%
	CLIP-FSAR (Wang et al., 2024)	CLIP-ViT-B	75.8%	87.7%	96.6%	99.0%
	MGTH	CLIP-ViT-B	80.5%	89.1%	97.5%	99.1%

Table 2

Comparison with state-of-the-art methods on the Kinetics and SSv2 datasets. The classification accuracy results for the 5-way 1-shot and 5-way 5-shot settings are illustrated. The symbol “-” denotes unavailable results in published works.

Modality	Method	Backbone	Kinetics		SSv2	
			1-shot	5-shot	1-shot	5-shot
Unimodal	OTAM (Cao et al., 2020)	ResNet-50	72.2%	84.2%	42.8%	52.3%
	TRX (Perrett et al., 2021)	ResNet-50	63.6%	85.2%	42.0%	64.6%
	HyRSM (Wang et al., 2022)	ResNet-50	73.7%	86.1%	54.3%	69.0%
	HyRSM++ (Wang et al., 2022)	ResNet-50	74.0%	86.4%	55.0%	69.8%
	ITANet (Zhang et al., 2021)	ResNet-50	73.6%	84.3%	49.2%	62.3%
	TADRNet (Wang et al., 2023)	ResNet-50	75.6%	87.4%	43.0%	61.1%
	STRM (Thatipelli et al., 2022)	ResNet-50	62.9%	86.4%	43.5%	68.1%
	HCL (Zheng et al., 2022)	ResNet-50	73.7%	85.8%	47.3%	64.9%
	GgHM (Xing et al., 2023)	ResNet-50	74.9%	87.4%	54.5%	69.2%
	MTFAN (Wu et al., 2022)	ResNet-50	74.6%	87.4%	45.7%	60.4%
	PAL (Zhu et al., 2021)	ResNet-50	74.2%	87.1%	46.4%	62.6%
	MTE (Jiang et al., 2026)	ResNet-50	72.3%	87.6%	49.4%	69.5%
	HM2 (Gao et al., 2025)	ResNet-50	75.5%	87.5%	-	-
	MSVT (Chen et al., 2025)	ResNet-50	77.2%	89.0%	41.0%	55.8%
Multimodal	AMeFu-Net (Fu et al., 2020)	ResNet-50	74.1%	86.8%	-	-
	CLIP-CPM ² C (Guo et al., 2025)	CLIP-ResNet50	88.6%	93.1%	58.0%	64.0%
	CLIP-MDMF (Guo et al., 2024)	CLIP-ResNet50	90.1%	93.5%	56.9%	68.0%
	MA-CLIP (Xing et al., 2025)	CLIP-ResNet50	92.8%	93.0%	58.8%	63.6%
	CLIP-FSAR (Wang et al., 2024)	CLIP-ResNet50	87.6%	91.9%	58.1%	62.8%
	MGTH	CLIP-ResNet50	88.1%	92.2%	60.2%	63.7%
	CLIP-CPM ² C (Guo et al., 2025)	CLIP-ViT-B	91.0%	95.5%	60.1%	72.8%
	CLIP-MDMF (Guo et al., 2024)	CLIP-ViT-B	95.0%	96.2%	60.1%	72.7%
	MA-CLIP (Xing et al., 2025)	CLIP-ViT-B	95.7%	96.0%	63.3%	72.3%
	CLIP-FSAR (Wang et al., 2024)	CLIP-ViT-B	89.7%	95.0%	61.9%	72.1%
	MGTH	CLIP-ViT-B	92.1%	96.3%	63.1%	73.9%

direction-sensitive (e.g., “Putting pen into cup” and “Taking pen out of cup”), and thus horizontal flipping is not applied. Subsequently, we adopt CLIP-ResNet50 and CLIP-ViT-B/16 as the embedding networks to extract 1024-dimensional and 512-dimensional features, respectively. The corresponding MGTH model configurations contain 47.1M and 90M

parameters. The entire training process spans 20 epochs, with each epoch consisting of 1,000 meta-tasks. Additionally, the model employs Adam for optimization, with an initial learning rate set to 0.0001. To further enhance training stability, we employ a Multi-Step Learning Rate Scheduler that decays the learning rate by a factor of 0.1 at the 5th, 10th,

15th, and 18th epochs, ensuring fast and stable convergence. For 5-way few-shot tasks, query samples are classified using the average prototype paradigm. The performance of the MGTH model is evaluated by computing the average accuracy across 10,000 tasks randomly sampled from the test set. All experiments are implemented in PyTorch, and training is conducted on two NVIDIA RTX4090Ti GPUs.

Performance comparison

Tables 1 and 2 showcase the superior performance of MGTH compared to state-of-the-art methods (Cao et al., 2020, Wu et al., 2022, Fu et al., 2020, Thatipelli et al., 2022, Wang et al., 2022, Perrett et al., 2021, Zhang et al., 2021, Huang et al., 2022, Nguyen et al., 2022, Wang et al., 2024, Wang et al., 2022, Wang et al., 2023, Zheng et al., 2022, Xing et al., 2023, Wang et al., 2024, Li et al., 2021, Zhu et al., 2021, Li et al., 2021, Liu et al., 2023, Wang et al., 2021, Jiang et al., 2026, Gao et al., 2025, Chen et al., 2025, Guo et al., 2025, Guo et al., 2024, Xing et al., 2025) on four few-shot action recognition datasets: Kinetics, UCF-101, HMDB-51, and SSv2. The experimental results highlight the following major observations:

- (1) Compared to unimodal approaches, our MGTH achieves optimal results across all datasets. For example, on the 5-way 1-shot setting on Kinetics and UCF-101, MGTH based on CLIP-ResNet50 surpasses the best unimodal method, MSVT (Chen et al., 2025), by considerable margins of 10.9% and 1.4%, respectively, while yielding comparable results on HMDB-51. Similarly, compared to HYRSM++ (Wang et al., 2022), which achieves state-of-the-art performance in RGB modality under the 1-shot SSv2 setting, MGTH improves performance from 55.0% to 60.2%. These notable advancements suggest that the complementarity between video and text modalities provides substantial benefits for the few-shot action recognition task. Furthermore, compared to widely used matching strategies (Cao et al., 2020, Perrett et al., 2021), MGTH outperforms OTAM (Cao et al., 2020) by 11.4% (63.7% vs. 52.3%) and slightly lags behind TRX (Perrett et al., 2021) by 0.9% (63.7% vs. 64.6%) on 5-shot SSv2, while remarkably surpassing both approaches on 1-shot SSv2. This divergence stems from the design of our framework, which is more generalized and not specifically tailored to any particular few-shot setting, unlike TRX (Perrett et al., 2021), which is optimized for the 5-shot task. Meanwhile, MGTH based on CLIP-ViT-B outperformed the CLIP-RN50-based approach across all task settings, demonstrating that more powerful pre-trained models yield superior few-shot generalization capabilities.
- (2) Compared to multimodal approaches, MGTH exhibits more significant advantages. More specifically, compared with AMeFuNet (Fu et al., 2020), which integrates depth information as a scene carrier with the RGB information, MGTH based on CLIP-ResNet50 achieves a notable improvement of 14% on the 1-shot setting of the Kinetics dataset. CCLN (Wang et al., 2024) employs a generative adversarial network to synthesize visual features and uses a video-text-based cross-modal contrastive learning module to acquire discriminative feature representations. The proposed MGTH outperforms CCLN by 14.2% (60.2% vs. 46.0%) and 2.4% (63.7% vs. 61.3%) in the few-shot setting on the SSv2 dataset. Compared with CLIP-CPM²C (Guo et al., 2025), CLIP-MDMF (Guo et al., 2024), and MA-CLIP (Xing et al., 2025), the MGTH model achieves performance gains of 7.6%, 7.1%, and 0.6%, respectively, on the HMDB-51 dataset under the 5-way 1-shot setting. Additionally, our MGTH surpasses the multimodal CLIP-FSAR (Wang et al., 2024), which adopts the temporal alignment strategy OTAM, by 4.7% (73.9% vs. 69.2%), 0.5% (91.8% vs. 91.3%), 0.5% (88.1% vs. 87.6%), and 2.1% (60.2% vs. 58.1%) in the 1-shot setting. It is noteworthy that although both CLIP-FSAR and our method utilize the CLIP framework for

Table 3

Impact of the proposed module on performance under the 5-way 1-shot setting.

CLIP-OTAM	Baseline (CLIP-TCH)	VTA	MTAM	TPCM	HMDB-51	SSv2
✓	×	×	×	×	63.0%	44.1%
×	✓	×	×	×	66.1%	47.5%
×	✓	×	✓	×	68.2%	49.8%
×	✓	✓	×	×	68.8%	50.5%
×	✓	×	×	✓	71.8%	57.4%
×	✓	✓	✓	✓	73.9%	60.2%

Table 4

Comparison results of different metric strategies on 5-way 1-shot task.

Metric strategy	Backbone	HMDB-51	SSv2
MGTH-OTAM	CLIP-ResNet50	71.3%	58.5%
MGTH-TRX	CLIP-ResNet50	70.7%	57.8%
MGTH-Bi-MHM	CLIP-ResNet50	71.6%	58.9%
MGTH-TCH (Ours)	CLIP-ResNet50	73.9%	60.2%

few-shot action recognition, MGTH employs several effective strategies to improve performance. Specifically, the MTAM module supplements visual information by incorporating motion features. The VTA extends the video-text contrastive objective of CLIP-FSAR and integrates three types of text prompt templates to jointly optimize the alignment between visual and textual features. Additionally, the TPCM module more effectively models relationships between different modalities, thereby fully utilizing textual information to supervise the generation of visual prototypes. Finally, compared to OTAM, the TCH provides a more flexible alignment strategy to achieve robust few-shot matching.

Ablation study

To evaluate the effectiveness of the four core modules in the MGTH model, we conducted ablation experiments on the 5-way 1-shot setting of the SSv2 and HMDB-51 datasets, as outlined in Table 3. We first use CLIP-ResNet50 as the visual encoder and apply OTAM for few-shot matching, resulting in the CLIP-OTAM network, which achieves accuracies of 63.0% and 44.1% on the HMDB-51 and SSv2 datasets, respectively. Next, with all other settings unchanged, we replaced OTAM with TCH as the baseline model (e.g., CLIP-TCH). The results indicate that the baseline model improves by 3.1% (63.0% vs. 66.1%) and 3.4% (44.1% vs. 47.5%) compared with CLIP-OTAM. This suggests that the TCH metric enhances the model's robustness to misaligned instances and its sensitivity to long-term temporal order, aiming to mitigate intra-class temporal offsets and inter-class local similarity. When combining MTAM with the baseline, the accuracy of SSv2 increases from 47.5% to 49.8%, underscoring the importance of motion and spatio-temporal features in action recognition. By incorporating the VTA module into the baseline, we obtain a 3% performance improvement on the SSv2 dataset, emphasizing the necessity of adapting CLIP for few-shot action recognition tasks. When TPCM is applied to the baseline, we observe accuracy increases by 9.9% and 5.7% on the SSv2 and HMDB-51 datasets, respectively. This can be attributed to the TPCM module's ability to efficiently utilize textual information to calibrate visual prototypes. Additionally, by integrating all modules, our method achieved the best performance, demonstrating the complementarity of different modules and confirming the rationality of our model architecture.

To validate the feasibility of the TCH strategy, the proposed MGTH method is incorporated into existing metrics, including OTAM, TRX, and Bi-MHM, as presented in Table 4. Experimental results prove that, compared to other matching methods, MGTH-TCH achieves optimal performance on the HMDB-51 and SSv2 datasets. This reveals that enforcing frame-level and tuple-level Bi-MHM during matching can

Table 5

Importance analysis of temporal coherence regularization strategy on SSv2 dataset.

Method	5-way 1-shot
Frame-level Bi-MHM	58.9%
Frame-level Bi-MHM + Temporal coherence regularization	59.5%
Tuple-level Bi-MHM	59.1%
Tuple-level Bi-MHM + Temporal coherence regularization	59.6%
Frame-level Bi-MHM+ Tuple-level Bi-MHM	59.5%
Frame-level Bi-MHM+ Tuple-level Bi-MHM+ Temporal coherence regularization (Ours)	60.2%

Table 6

Effect of different prototype calibrations on 5-way 1-shot tasks.

Prototype calibration	HMDB-51	Kinetics
Bi-LSTM	64.2%	81.6%
Bi-GRU	64.5%	82.2%
Multi-head Self-Attention Mechanism	73.1%	87.5%
Meta-fusion module (Han et al., 2024)	73.4%	87.7%
TPCM (Ours)	73.9%	88.1%

effectively mitigate intra-class temporal offsets and inter-class local similarity. Meanwhile, imposing temporal coherence regularization on the input video facilitates the capture of long-range temporal order cues and thus enhances the effectiveness of the matching strategy.

We conduct six comparative experiments on the 5-way 1-shot setting of the SSv2 dataset to demonstrate the superiority of temporal coherence regularization, as presented in Table 5. The results confirm that incorporating temporal coherence regularization into Frame-level Bi-MHM, Tuple-level Bi-MHM, or Frame-level Bi-MHM + Tuple-level Bi-MHM significantly improves model performance. This is attributed to the temporal coherence regularization, which pulls adjacent frame features closer while pushing non-adjacent frames apart, thereby enhancing the model's ability to capture long-term temporal order for more accurate matching. Furthermore, compared to Frame-level Bi-MHM and Tuple-level Bi-MHM, the Frame-level Bi-MHM + Tuple-level Bi-MHM yields accuracy improvements of 0.6% and 0.4% on the SSv2 dataset, respectively. This demonstrates the effectiveness of the Frame-level Bi-MHM+ Tuple-level Bi-MHM approach in handling misalignment instances.

To ensure a fair comparison, we replace the TPCM component in the MGTH model with a Bi-LSTM, a Bi-GRU, a multi-head self-attention mechanism, and a meta-fusion architecture, respectively, to investigate the impact of different prototype calibration strategies on model performance, as presented in Table 6. In these alternative configurations, the extracted textual features are first concatenated with the corresponding support visual features along the temporal dimension, and then fed into the Bi-LSTM, Bi-GRU, or multi-head self-attention module to achieve cross-modal integration. It is worth noting that the Meta-fusion architecture, proposed in (Han et al., 2024), employs a multi-head cross-attention mechanism that uses textual features as query and visual features as key and value to mitigate the sparsity of few-shot visual representations. Although this method is conceptually related to the present study, our TPCM offers a more effective mechanism for capturing the coupling between textual semantics and visual

Table 7

Comparison results of different prompt templates on the HMDB-51 and SSv2 datasets.

Prompt template	HMDB-51	SSv2
[CLS]	71.7%	59.2%
a photo of a [CLS]	72.0%	59.1%
Prefix prompt: [CLS], a video of action	73.9%	60.2%
Cloze prompt: this is [CLS], a video of action		
Suffix prompt: human action of [CLS]		

Table 8

Comparison of motion modeling with and without 1D Convolution on the 5-way 1-shot task.

Motion representation	HMDB-51	UCF-101	SSv2
No-1D Conv	73.5%	91.3%	59.4%
1D Conv (both)	73.6%	91.3%	59.7%
1D Conv (after)	73.9%	91.8%	60.2%

Table 9

Comparison of spatiotemporal features and motion features under the 5-way 1-shot setting.

Model	HMDB-51	SSv2
MTAM-M	72.8%	57.3%
MTAM-T	73.3%	58.8%
MTAM(Ours)	73.9%	60.2%

features, thereby yielding more discriminative class-prototype representations. The experimental results further demonstrate that TPCM delivers notable advantages in cross-modal alignment and feature calibration.

In the proposed MGTH, we explore the impact of different text prompts on performance, as displayed in Table 7. Unlike the traditional prompt “a photo of a [CLS]” and “[CLS]”, we employ three text prompt templates: Prefix prompt, Cloze prompt, and Suffix prompt. These templates are designed to enhance the extraction of critical video features, facilitate video-text alignment, and thus strengthen cross-modal integration. Experimental results on the HMDB-51 and SSv2 datasets demonstrate that these templates significantly improve performance, further validating the applicability of our approach.

To assess the importance of the 1D convolution in Eq. (6), we conducted comparative experiments on the UCF-101, HMDB-51, and SSv2 datasets, as summarized in Table 8. In this ablation, “1D Conv (both)” denotes applying the 1D convolution to both adjacent frames, whereas “1D Conv (after)” refers to applying it only to the later frame, which corresponds to our proposed design. The results demonstrate that incorporating the 1D convolution consistently improves model performance across all datasets, confirming the effectiveness of this component. This improvement arises from the fact that directly differencing features in their original space tends to amplify task-irrelevant noise channels while suppressing channels critical to motion semantics. Although applying the convolution to both frames yields some performance gain, the improvement is noticeably smaller. This is primarily because performing identical convolutions on adjacent frames produces highly similar feature responses, thereby weakening the representation of motion cues. In contrast, applying the 1D convolution only to the subsequent frame enables the model to learn channel representations that better capture temporal changes, ultimately enhancing the discriminative power of motion features.

To further analyze the respective contributions of spatiotemporal features and motion features within the MTAM module, we conducted a series of ablation experiments, as outlined in Table 9. Among these, “MTAM-T” denotes retaining only the Temporal Transformer to model spatiotemporal dependencies. In other words, this component is designed to capture correlations between spatial appearance features across different time points. Taking the “long jump” as an example, videos typically begin with the athlete's approach run, with the sandpit appearing only at the end of the footage. If the model fails to incorporate the semantic information of the sandpit into the spatial appearance features of the preceding frames, it is prone to misclassifying this action as “running.” In contrast, “MTAM-M” relies solely on the difference between adjacent frames to learn feature-level motion information. Without adequate motion cues, the model struggles to distinguish actions with similar trajectories but opposite directions—such as “pulling a plug” versus “inserting a plug.” The experimental results show that

Table 10

Effectiveness analysis of the MTAM module.

Model	Backbone	FLOPs (G)	Param (M)	Acc (%)
OTAM (Cao et al., 2020)	ResNet-50	8.17	23.5	42.8
TRX (Perrett et al., 2021)	ResNet-50	8.22	47.1	42.0
MGTH-C3D	ResNet-50	9.80	55.6	52.3
MGTH-MTAM	ResNet-50	8.26	38.2	56.7

Table 11

The Impact of different cardinalities on model performance.

Ω	Num tuples	Kinetics	SSv2
{3}	56	87.8%	59.8%
{4}	70	87.2%	60.5%
{2} (Ours)	28	88.1%	60.2%

neither spatiotemporal features nor motion features alone can achieve optimal performance. This observation indicates that the two types of information are highly complementary in video action recognition.

To provide a more comprehensive understanding of the MTAM module, Table 10 presents a detailed comparative analysis of MGTH against multiple methods in terms of FLOPs, parameter scale, and classification accuracy on the 5-way 1-shot task of the SSv2 dataset. In this experiment, we replace MTAM with a C3D block to form the MGTH-C3D model, while keeping all other components identical to ensure a fair comparison. The experimental results reveal that MTAM reduces FLOPs by approximately 15.7% and parameters by about 31.3% compared with MGTH-C3D. This substantially decreases the model's computational and storage costs and demonstrates the rationality of the proposed design.

Table 11 illustrates the effect of different cardinalities on temporal relationship modeling during the hybrid matching process. We conduct 5-way 1-shot experiments on both the Kinetics and SSv2 datasets to evaluate the performance of cardinality settings $\Omega \in \{2, 3, 4\}$. For the spatially oriented Kinetics dataset, $\Omega = \{2\}$ achieves the highest accuracy of 88.1%, suggesting that the model focuses more effectively on background cues during matching. In contrast, for the more temporally dependent SSv2 dataset, using $\Omega = \{4\}$ further strengthens temporal relationship modeling and yields the best performance of 60.5%. However, increasing the cardinality also enlarges the number of tuples, which leads to a substantial rise in computational cost and inference time. For example, $\Omega = \{4\}$ generates more than twice as many tuples as $\Omega = \{2\}$, while offering only marginal improvements in recognition performance. Therefore, considering both performance and computational efficiency, we adopt $\Omega = \{2\}$ as the default configuration in this study.

Table 12 illustrates the effect of different β values on model performance under the 5-way 1-shot setting on the Kinetics and SSv2 datasets. The parameter β controls the weight of the temporal coherence regularization in the final loss function. The experimental results indicate that on the Kinetics dataset, MGTH achieves the best performance when $\beta = 0.3$, corresponding to a weight ratio of 1: 0.3: 0.35: 0.35 for the loss terms L_{hybrid} , $L_{\text{temporal-regular}}$, $L_{\text{video-text}}$ and $L_{\text{text-video}}$. In contrast, on the SSv2 dataset, the optimal value is $\beta = 0.4$, with a corresponding weight ratio of 1: 0.4: 0.3: 0.3. This suggests that temporal coherence regularization plays a more prominent role on the SSv2 dataset. This is because SSv2 requires stronger temporal reasoning, which allows MGTH to

Table 12Sensitivity analysis of the hyperparameter β .

Param β	SSv2	Kinetics
0.1	58.4%	87.5%
0.2	58.8%	87.8%
0.3	60.0%	88.1%
0.4	60.2%	87.7%
0.5	58.6%	87.5%

Table 13

Performance comparison of different cross-domain few-shot action recognition methods when using Kinetics100-Small as the source-domain dataset.

Method	HMDB-51	UCF-101	SSv2
TRX (Perrett et al., 2021)	28.2%	44.5%	22.8%
STRM (Thatipelli et al., 2022)	27.5%	45.0%	23.1%
HySRM (Wang et al., 2022)	32.1%	56.2%	24.6%
SEEN (Wang et al., 2023)	39.5%	64.6%	26.2%
DMSD (Guo et al., 2025)	39.6%	65.0%	26.7%
MGTH	45.8%	72.2%	35.9%

Table 14

Performance comparison with zero-shot action recognition methods.

Method	HMDB-51(%)	UCF-101(%)
ActionCLIP (Wang et al., 2023)	40.8	58.3
Vita-CLIP (Wasim et al., 2023)	48.6	75.0
XCLIP (Ni et al., 2022)	44.6	72.0
ViFi-CLIP (Rasheed et al., 2023)	53.9	76.2
ViFi-CLIP + Hierarchical Adapter (Ge et al., 2025)	54.9	77.6
MGTH	53.8	76.0

achieve more substantial performance gains on this dataset.

Extension to cross-domain few-shot action recognition

Table 13 To further validate the applicability of the proposed MGTH method in cross-domain scenarios, we extend it to cross-domain few-shot learning tasks to mitigate the performance degradation caused by distribution shifts between the source and target domains. To ensure comparability with existing studies, we follow the experimental setup in [79], adopting ResNet-18 as the backbone and removing its final fully connected layer to obtain 512-dimensional frame-level features. Additionally, we employ Kinetics-Small as the source-domain dataset and perform cross-domain experiments on UCF-101, HMDB-51, and SSv2 as target domains. Table 14 presents the performance comparison of various methods across the above cross-domain combinations. We observe that MGTH achieves the best performance across all cross-domain tasks. Compared with SEEN (Wang et al., 2023), MGTH improves 1-shot accuracy on UCF-101 from 64.6% to 72.2%, on HMDB-51 from 39.5% to 45.8%, and on SSv2 from 26.2% to 35.9%. These results clearly demonstrate the effectiveness and strong generalization capability of MGTH in cross-domain few-shot action recognition.

Extension to zero-shot action recognition

In this section, we further investigate the effectiveness of MGTH in zero-shot action recognition. Following the setup of existing zero-shot video recognition work (Ge et al., 2025), we use the pretrained CLIP ViT-B/16 as the backbone for MGTH and adopt Top-1 accuracy as the evaluation metric. Table 14 illustrates the comparative performance of MGTH and several zero-shot methods on the HMDB-51 and UCF-101 datasets. The results show that MGTH achieves accuracies of 53.8% and 76.3%, respectively, substantially outperforming existing methods such as ActionCLIP (Wang et al., 2023), Vita-CLIP (Wasim et al., 2023), and XCLIP (Ni et al., 2022), while performing comparably to ViFi-CLIP (Rasheed et al., 2023). However, a performance gap remains when compared with the state-of-the-art ViFi-CLIP + Hierarchical Adapter (Ge et al., 2025). This discrepancy is primarily attributed to differences in motion information modeling between the two approaches. MGTH employs only simple adjacent frame difference operations to obtain motion representations, whereas ViFi-CLIP + Hierarchical Adapter (Ge et al., 2025) leverages metrics based on predictive entropy and temporal probability difference to assess the model's confidence in single-frame predictions and its sensitivity to temporal discriminative cues, thereby capturing more reliable motion information. Overall, these findings

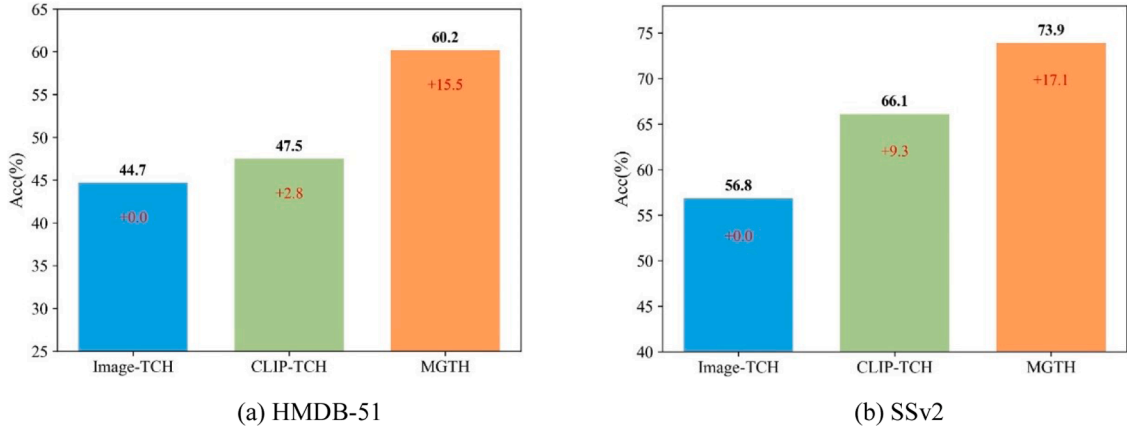


Fig. 5. Performance comparison of different methods under the 5-way 1-shot setting on the HMDB-51 and SSv2 datasets.

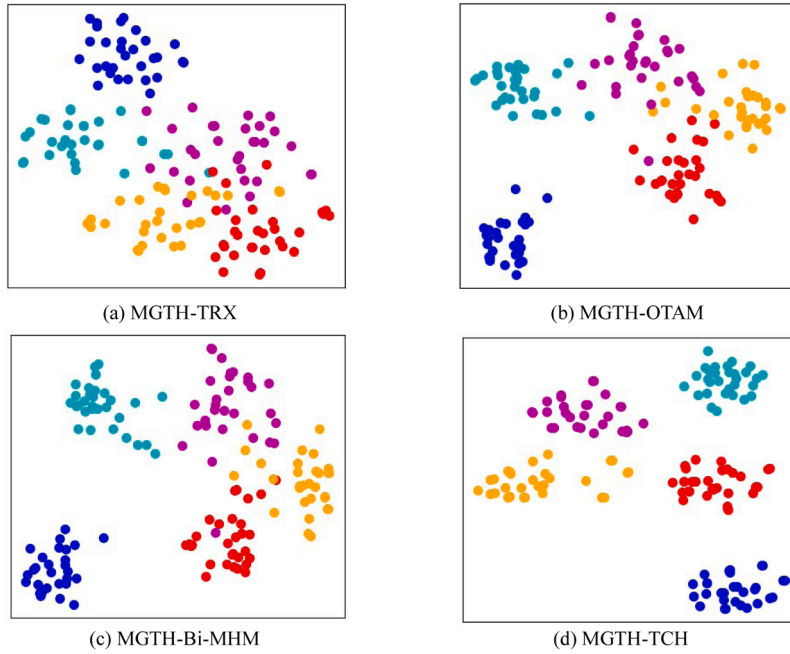


Fig. 6. Visualization of the T-SNE feature distribution for five action classes on the HMDB-51 dataset. These five classes are “brush hair”, “chew”, “clap”, “draw_sword” and “dribble”.

suggest that MGTH exhibits promising potential in zero-shot action recognition. In future work, we plan to further explore the rich textual semantics provided by large language models and incorporate cross-modal semantic alignment strategies, such as those introduced in (Xie et al., 2019, Xie et al., 2025), to enhance the scalability of MGTH in zero-shot scenarios.

Visualization analysis

We comprehensively analyze the multimodal properties of the MGTH model, with experimental results summarized in Fig. 5. Here, “Image-TCH” indicates that the visual features output from the ResNet-50 network pre-trained on ImageNet are directly input into the TCH matching strategy. To transfer CLIP’s strong generalization capability to the few-shot action recognition task, we adopt its pre-trained visual encoder to replace the encoder of Image-TCH, forming “CLIP-TCH,” which is then fine-tuned. The experimental results indicate that, compared to Image-TCH, CLIP-TCH improves accuracy by 9.3% and 2.8% on the HMDB-51 and SSv2 datasets, respectively. In contrast, on

these two datasets, the MGTH model achieves at least a 15.5% improvement over the Image-TCH and a minimum of 7.8% over the CLIP-TCH approach. This demonstrates that the proposed approach fully leverages multimodal video-text information to enhance the representativeness of video class prototypes and dramatically improve CLIP’s adaptability in few-shot action recognition tasks.

To further demonstrate the superiority of the proposed TCH, we analyze the feature distribution variations of different matching methods on the HMDB-51 dataset, as illustrated in Fig. 6. By keeping all other settings unchanged, we replaced the TCH matching strategy in the MGTH model with TRX, OTAM, and Bi-MHM, respectively. The experimental results reveal that MGTH-TCH significantly improves the compactness of intra-class distributions and the separation of inter-class distributions compared to MGTH-TRX, MGTH-OTAM, and MGTH-Bi-MHM. For instance, in Fig. 5(d), the ambiguity among the categories “brush hair,” “chew,” and “dribble” is noticeably reduced.

This improvement arises from TCH’s ability to alleviate intra-class temporal offsets and inter-class local similarity while fully exploiting the smoothness and continuous changes of the video.

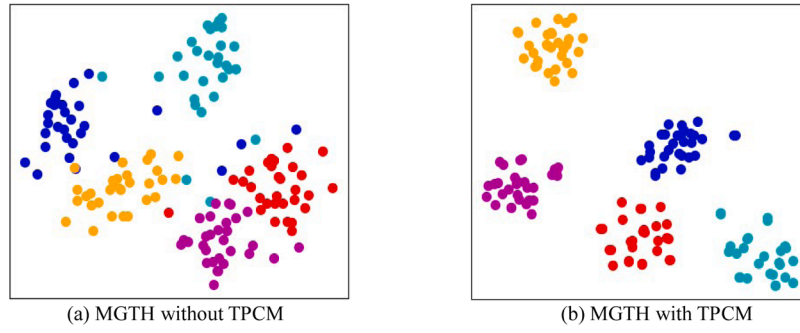


Fig. 7. Visualization of the T-SNE feature distribution for five action categories on the SSv2 dataset. These five classes are “Approaching something with our camera”, “Taking something out of something”, “Pushing something from right to left”, “Pouring something out of something” and “Showing something next to something”.

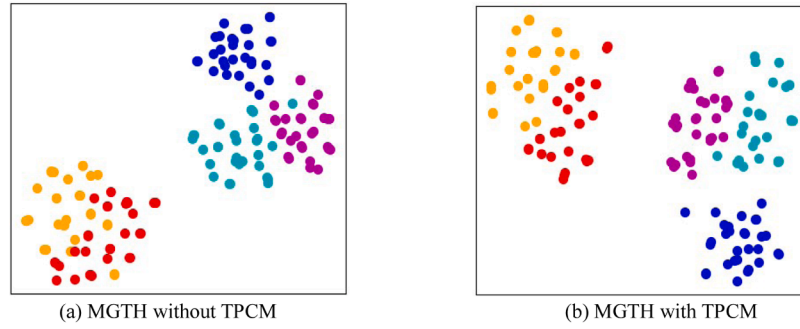


Fig. 8. Visualization of the T-SNE feature distribution for five action categories on the UCF-101. These five classes are “PlayingCello”, “PlayingSitar”, “PlayingGuitar”, “JumpingJack” and “JumRope”.

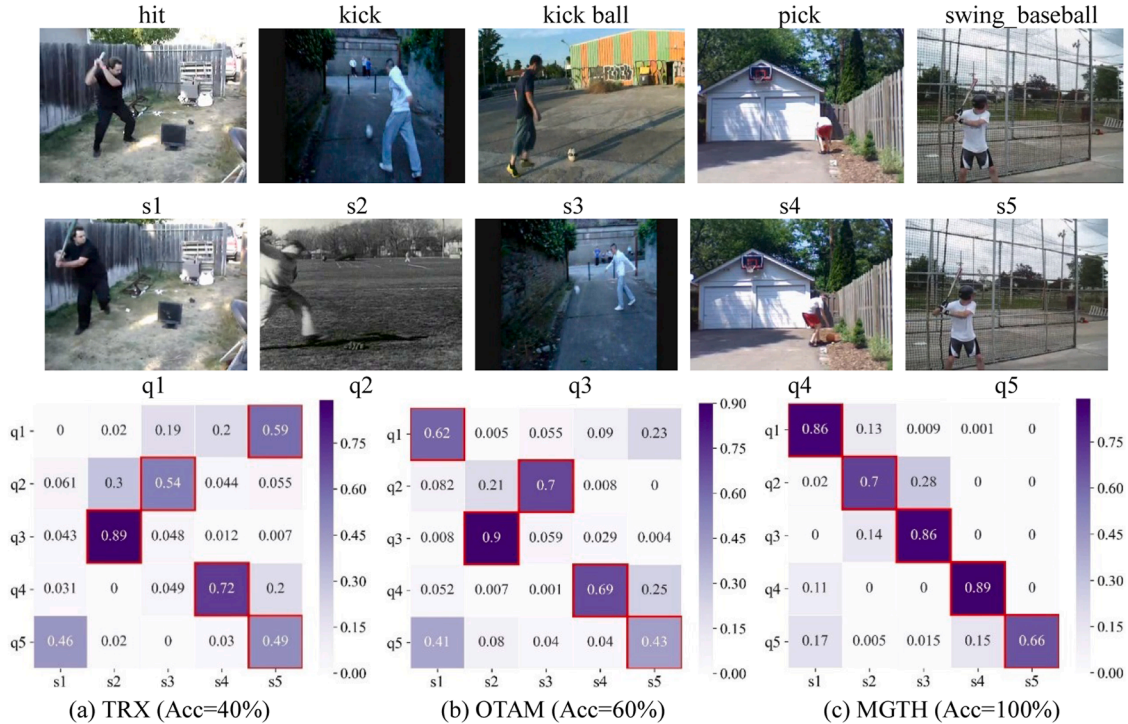


Fig. 9. Visualization of the similarities relationship between query and support videos under the 5-way 1-shot setting on the HMDB-51 dataset. Higher scores represent greater similarity.

To visually highlight the contributions of the proposed prototype calibration method, we visualize the feature distributions of the SSv2 dataset with and without TPCM, as presented in Fig. 7. The comparison results reveal that TPCM improves the inter-class discrimination and

narrows the intra-class difference. For example, in Fig. 7(b), the two fine-grained action classes, “Taking something out of something” and “Pushing something from right to left,” are clearly distinguished from each other. Additionally, we extend this analysis to the UCF-101 dataset,

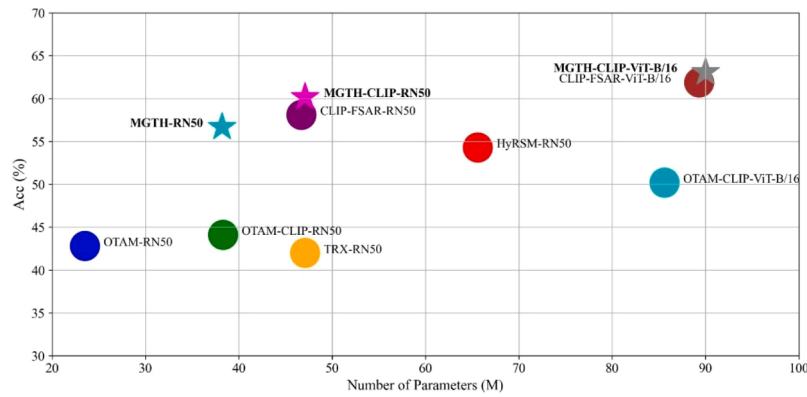


Fig. 10. Performance comparison of various methods in the 5-way 1-shot setting of the SSv2 dataset. The horizontal coordinate of the center point of each star or circle represents the parameter count, while the vertical coordinate represents the recognition accuracy.

as shown in Fig. 8. These experiments demonstrate that MGTH with TPCM adequately employs CLIP’s rich textual semantic priors to calibrate prototypes, leading to more compact intra-class distributions and greater inter-class distinctions.

To qualitatively assess MGTH’s ability to extract task-specific discriminative patterns, Fig. 9 visualizes the similarity between query and support videos across different approaches. Importantly, the actions “hit” and “swing_baseball” exhibit similar swinging motions yet belong to different classes. Similarly, “kick” and “kick ball” share overlapping visual features, making them difficult to distinguish. Experimental results reveal that OTAM (Bishay et al., 2019) and TRX (Dhiman and Vishwakarma, 2019) struggle to handle categories with similar motion patterns. In contrast, MGTH effectively distinguishes these four fine-grained categories, aligning with our motivation. This demonstrates that the proposed method leverages motion features and CLIP’s powerful multimodal knowledge to compensate for the sparsity of few-shot spatiotemporal representations. Furthermore, the TCH matching strategy further improves the recognition performance for visually similar categories.

To gain further insight into MGTH, Fig. 10 illustrates its advantages over other methods in terms of parameters and accuracy. In this experiment, “MGTH-RN50” denotes the configuration where ResNet-50 pre-trained on ImageNet is employed as the visual encoder, while the text encoder remains unchanged. When using CLIP-RN50 as the visual encoder, MGTH-CLIP-RN50 achieves the best result, reaching 60.2% among models based on CLIP-RN50 and RN50, while maintaining a parameter count of 47.1M—lower than that of HyRSM and comparable to TRX. Furthermore, with ViT-B/16 as the visual encoder, MGTH-CLIP-ViT-B/16 attains the highest accuracy across all methods. Notably, it improves accuracy from 61.9% to 63.1% compared to CLIP-FSAR-ViT-B/16, with only a slight parameter increase from 89.3M to 90.0M. These results demonstrate that MGTH delivers substantial performance gains with minimal additional parameters.

Additionally, we perform a statistical evaluation of the performance improvement across categories relative to CLIP-TCH and MGTH-OTAM, as illustrated in Fig. 11. Based on the results, it can be observed that the proposed MGTH significantly improves the accuracy across all categories. Notably, in Fig. 11(a), categories such as “Dropping something next to something,” “Tipping something over,” and “Taking something out of something” exhibit accuracy gains of at least 24%, with similar improvements observed in Fig. 11(b). The excellent performance can be attributed to the optimization strategy proposed by MGTH, including MTAM for encoding complementary spatiotemporal and motion features, VTA for enhancing the adaptability of CLIP on a few-shot action

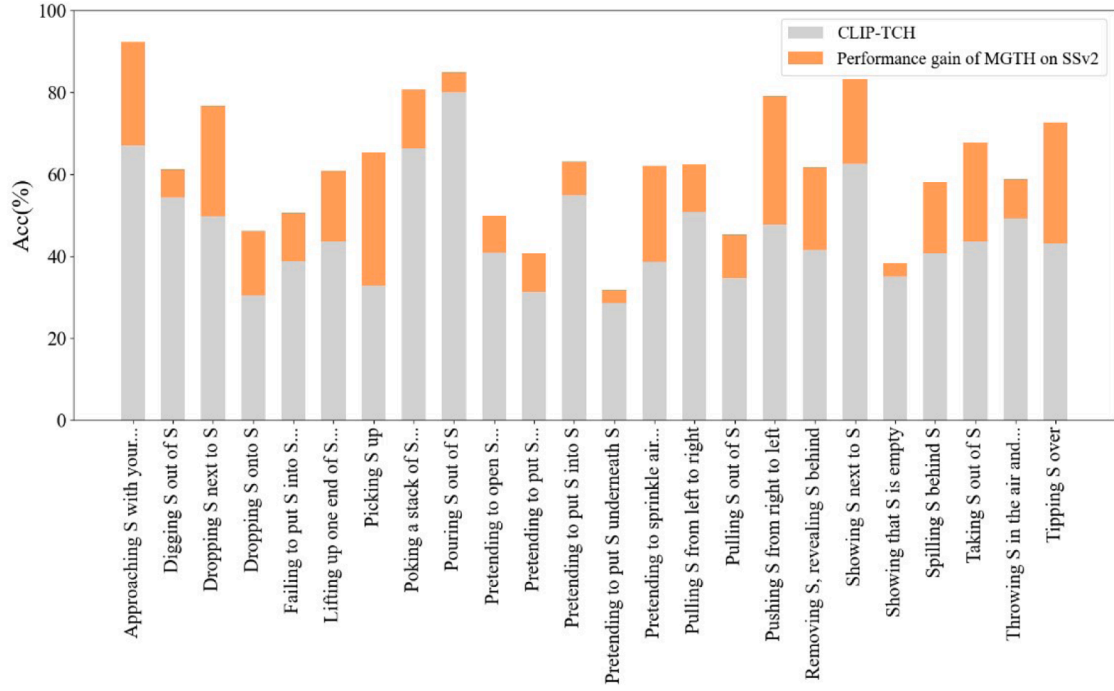
recognition task, TPCM for prototype calibration, and TCH for robust frame matching. The incorporation of these strategies enables MGTH to achieve better recognition rates and stronger generalization in few-shot action recognition.

Limitations

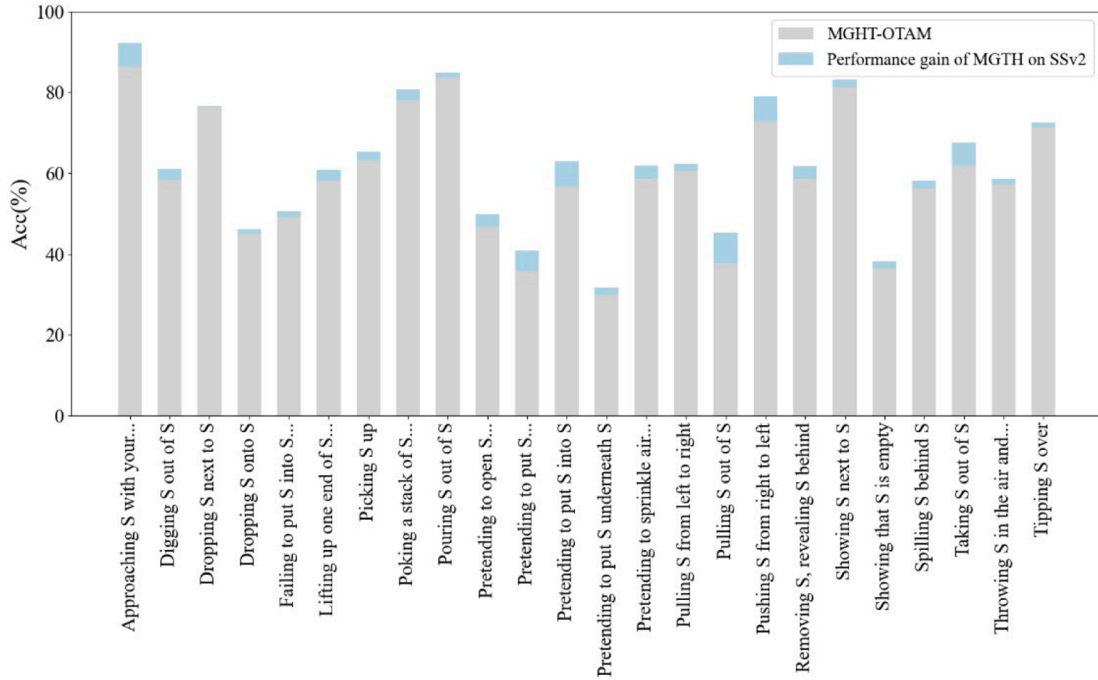
Although the proposed MGTH achieves notable performance improvements in few-shot action recognition, its architectural design inevitably leads to increased FLOPs and longer inference time, as shown in Table 15. This is primarily attributed to the incorporation of multi-head self-attention, cross-modal cross-attention, and the hybrid matching module composed of frame-level and tuple-level components within the model. Consequently, future work will focus on developing more efficient architectures and lightweight designs to achieve a better trade-off between performance and computational cost. Furthermore, compared with the commonly used CLIP prompt template “a photo of [CLS],” the prompts constructed in this paper—“[CLS], a video of action,” “this is [CLS], a video of action,” and “human action of [CLS]”—introduce greater semantic diversity. However, these prompts remain manually designed static templates, making them less adaptable to different categories and scenarios, thereby limiting further improvements in model performance. To address this limitation, we intend to incorporate learnable prompts in future research. For example, CoOp (Zhou et al., 2022) replaces the context words in a prompt with a set of learnable vectors and learns effectively with only a small number of labeled images. While the context learned by CoOp exhibits strong discriminative ability for base categories such as “arrival gate” and “cathedral,” its performance drops significantly when transferred to novel categories like “wind farm” or “train railway,” even though the task’s nature remains unchanged. This indicates that CoOp tends to overfit the base classes observed during training. Motivated by this consideration, future work may explore conditional prompt learning, which dynamically adapts the prompts to each instance and thus improves generalization to novel categories while mitigating overfitting.

Conclusion

In this work, we propose a multimodal-guided prototype calibration and temporal coherence-aware hybrid matching (MGTH) framework that achieves remarkable progress in distinguishing similar classes. Our approach consists of four key components: the motion-enhanced temporal aggregation module (MTAM) models complementary spatiotemporal and motion representations without requiring 3D convolutions or



(a) Performance comparison of CLIP-TCH and MGTH on SSv2.



(b) Performance comparison of MGTH-OTAM and MGTH on SSv2.

Fig. 11. Impact of CLIP-TCH, MGTH-OTAM, and MGTH on per-class accuracy under the 5-way 1-shot setting on the SSv2 test dataset. “S” is the abbreviation of “something”.

optical flow computations; the text-guided prototype calibration module (TPCM) utilizes text semantic priors from CLIP to adaptively refine visual prototypes; the video-text adapter objective (VTA) bridges the gap between CLIP and few-shot video tasks by pulling the visual features closer to the corresponding category text representations; and the TCH employs frame-level and tuple-level bidirectional Mean Hausdorff Metric (Bi-MHM) to mitigate metric bias caused by intra-class temporal offsets and inter-class local similarity, and incorporates temporal

coherence regularization to better handle diverse few-shot video tasks. Moreover, we further extend MGTH to address the more challenging tasks of cross-domain few-shot action recognition and zero-shot action recognition. Experimental results demonstrate that our MGTH outperforms competitive baselines under various few-shot settings on four challenging datasets, underscoring its potential as a significant addition to the field.

Table 15

Complexity analysis under the 5-way 1-shot setting on SSv2.

Model	Backbone	FLOPs (G)	Latency (ms)	Acc (%)
OTAM (Cao et al., 2020)	ResNet-50	8.17	123.6	42.8
TRX (Perrett et al., 2021)	ResNet-50	8.22	96.9	42.0
HyRSM (Wang et al., 2022)	ResNet-50	8.36	87.8	54.3
MGTH	ResNet-50	8.26	99.1	56.7
CLIP-FSAR (Wang et al., 2024)	CLIP-ResNet50	11.6	102.3	58.1
MGTH	CLIP-ResNet50	11.9	105.6	60.2
CLIP-FSAR (Wang et al., 2024)	CLIP-ViT-B	33.7	128.4	61.9
MGTH	CLIP-ViT-B	34.2	133.7	63.1

CRedit authorship contribution statement

Yiyuan An: Writing – original draft, Software, Methodology, Conceptualization. **Yingmin Yi:** Writing – review & editing, Validation, Methodology, Formal analysis. **Yiwei Yuan:** Supervision, Formal analysis. **Rui Yu:** Investigation, Formal analysis. **Qiming Xue:** Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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